

# SMART INDIA HACKATHON 2026



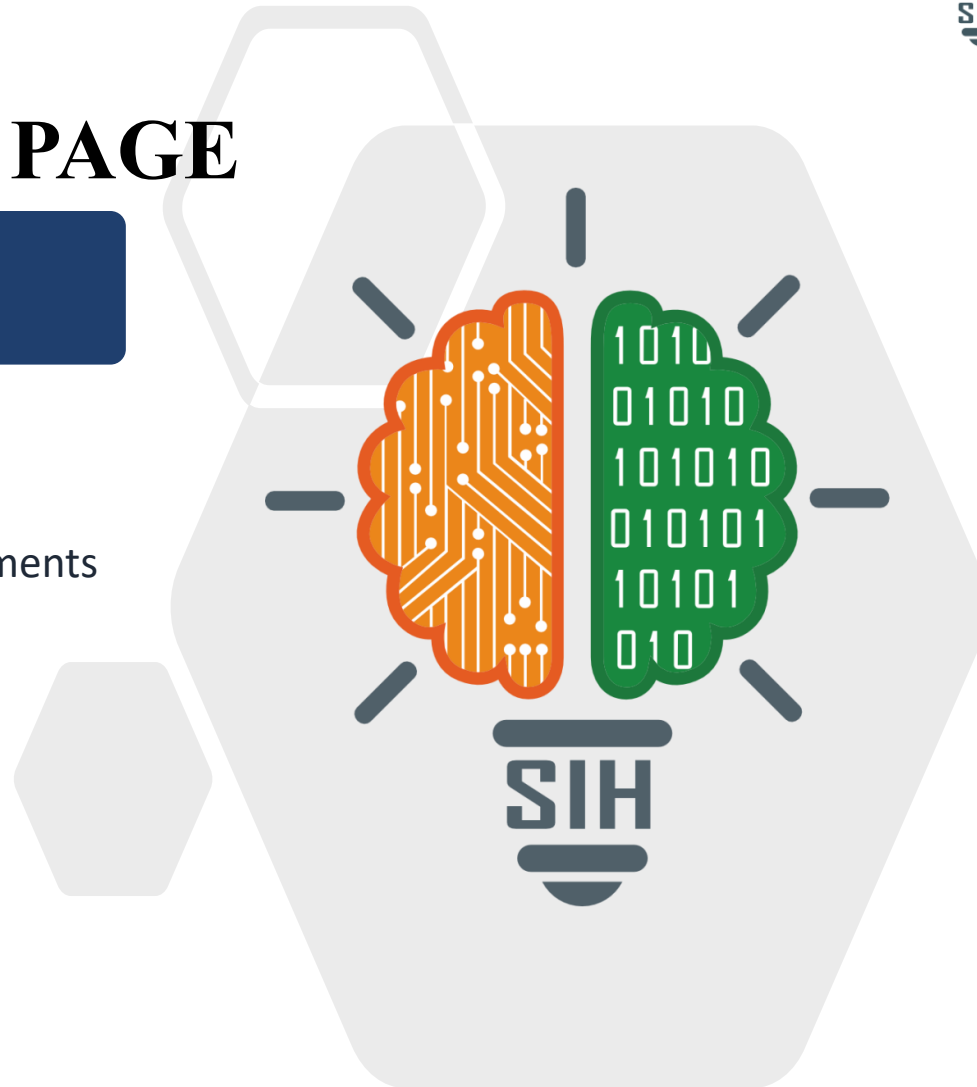
## TITLE PAGE

**TULA** Unified Digital Lifecycle Platform for Legal Metrology

- **Problem Statement ID** – 26036
- **Problem Statement Title** – Development of an Online Verification System for Weighing and Measuring Instruments
- **Theme** – Miscellaneous
- **PS Category** – Software
- **Team ID** – 182718
- **Team Name** – FriendlyFire

*One instrument → one digital ID → one certificate anyone can check*

**Live prototype:** [tula-legal-metrology.vercel.app](https://tula-legal-metrology.vercel.app)





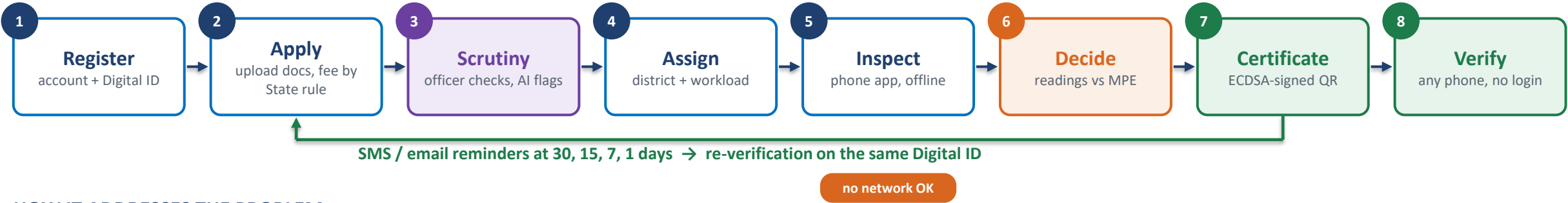
# TULA: ONE INSTRUMENT, ONE DIGITAL ID



## PROPOSED SOLUTION

**A live web + mobile platform: every instrument gets a permanent Digital ID, and application, scrutiny, field inspection (works offline), signed QR certificate, reminders and enforcement all attach to it.**

## DETAILED EXPLANATION: the life of one instrument (built and running)



## HOW IT ADDRESSES THE PROBLEM

Paper records in isolated systems  
→ **One live database for every State**

Manual scheduling, pendency  
→ **Enforced workflow + pendency dashboard**

Fake or copied certificates  
→ **Signed QR: edited copies fail the scan**

Missed re-verification dates  
→ **Automatic SMS / email reminders**

## INNOVATION AND UNIQUENESS

**Unfakeable QR**  
Server-signed (ECDSA P-256). Checked on any phone, even offline, plus a live revocation list

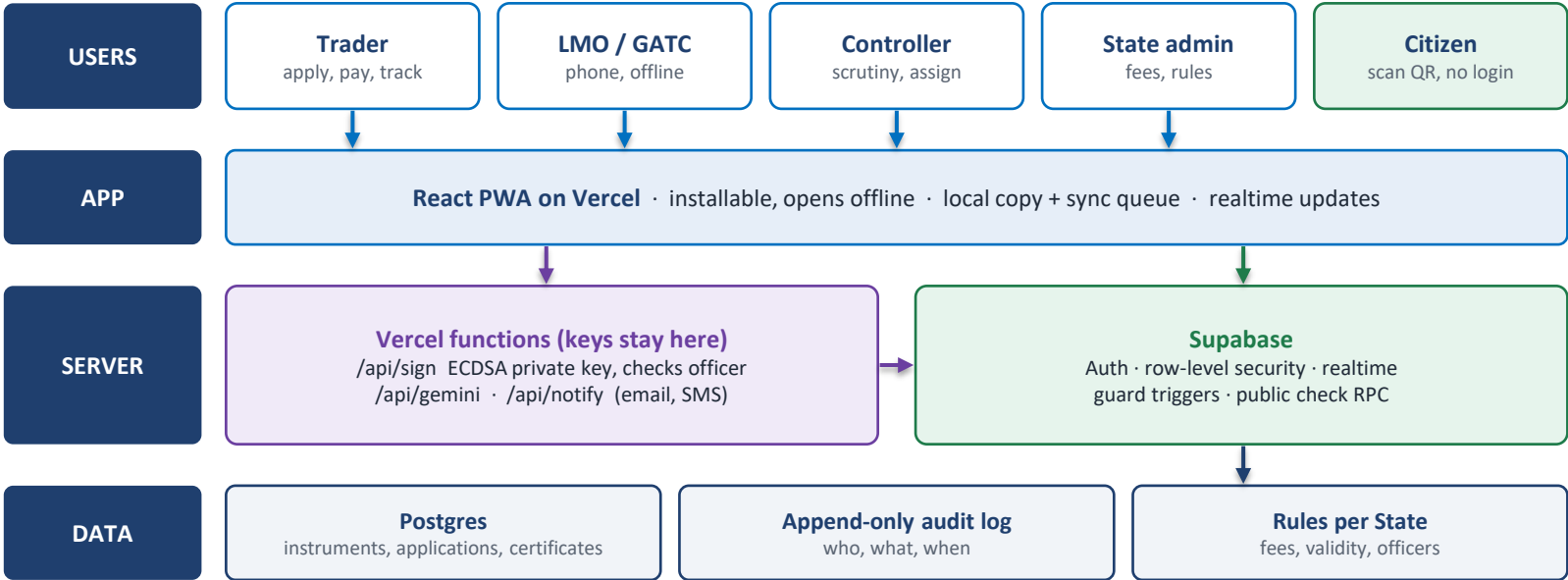
**Offline field app**  
Installable app. Inspection saves on the phone and syncs, signs and notifies on reconnect

**Computed verdict**  
Readings are checked against OIML MPE limits. PASS is impossible unless every check passes

**Multi-State by design**  
Fees, validity and officers are data per State. Database rules isolate each State

**AI copilot, human decides**  
Gemini flags odd documents. It never approves, prices or passes anything

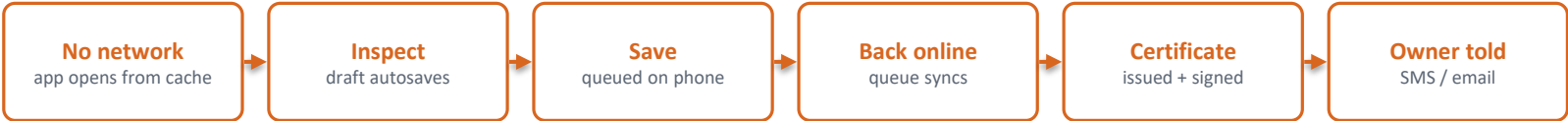
## METHODOLOGY AND PROCESS FOR IMPLEMENTATION (as deployed)



## HOW A FAKE QR IS CAUGHT



## OFFLINE FIELD INSPECTION



## TECHNOLOGIES USED

- React 19 · TypeScript · Vite · Tailwind installable, mobile-first
- Service worker · local store · sync queue offline work, auto sync
- Supabase Postgres · RLS · Auth · Realtime live, per State
- Vercel functions · ECDSA P-256 · jsQR sign on server, check on phone
- Google Gemini (Flash) document flags, help chat
- Resend · Twilio · jsPDF email, SMS, certificate PDF

## ON A PHONE, FROM THE LIVE PROTOTYPE

Genuine

Edited copy

Expired

Revoked

"Edited copy" takes the genuine certificate and changes only its expiry date. The signature no longer matches, so the check fails.

🚨 Fake or edited certificate

The signature does not match the certificate details. It was edited or forged. It claims to be DL/LM/2026/08912. Report it so the district office can inspect.

Weight or reading looks wrong, seal broken, or certificate expired?

Report a problem

Buyer scans an edited copy: rejected

Zero load

PASS

Test load

0.000 kg

Error

0.000 kg

Reading (kg)

0.000

Limit (MPE)

± 0.003 kg

25 % of Max

FAIL

Test load

7.500 kg

Error

+0.012 kg

Reading (kg)

7.512

Limit (MPE)

± 0.005 kg

50 % of Max

PASS

Test load

15.000 kg

Error

0.000 kg

Reading (kg)

15.000

Limit (MPE)

± 0.008 kg

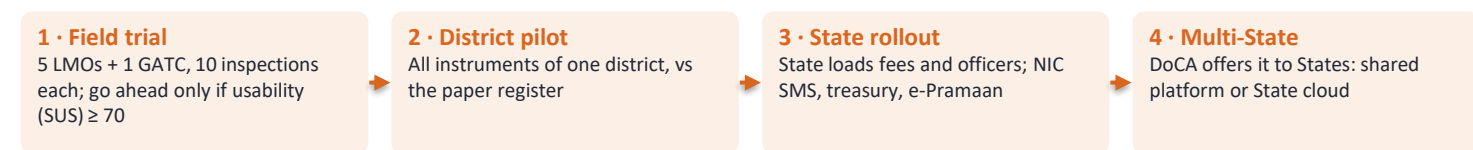
Over the legal limit: flagged at once



## ANALYSIS OF THE FEASIBILITY OF THE IDEA: what is built vs planned

| PS requirement                       | Live today                                  | Next                  |
|--------------------------------------|---------------------------------------------|-----------------------|
| Registration, profiles, secure login | ✓ real accounts, 6 roles                    | e-Praamaan, phone OTP |
| Apply for (re-)verification          | ✓ docs, fee, approval mark vs DoCA register | treasury payment      |
| Scheduling, allocation LMO / GATC    | ✓ district + workload                       | officer calendars     |
| Inspection results recorded          | ✓ MPE computed, photos, GPS                 | cloud photo storage   |
| QR certificate + authentication      | ✓ signed QR, revocation                     | HSM-held key          |
| Validity tracking, alerts            | ✓ 30/15/7/1-day reminders                   | nightly server job    |
| SMS / email updates                  | 🕒 email + SMS wired, keys pending           | NIC SMS gateway       |
| Dashboards: pending, inspections     | ✓ stage, ageing, district                   | state-wide wallboard  |
| Search, export, print                | ✓ search, PDF, CSV                          | bulk audit export     |
| Mobile app for field officers        | ✓ installable, offline                      | Android package       |

## HOW WE TEST WITH REAL OFFICERS, THEN SCALE



## VIABILITY: HOW IT GROWS, SURVIVES AND IS FUNDED

**Grow in the market**  
Buyers are DoCA and the 36 State / UT Legal Metrology departments; traders and citizens use it free. Each State that adopts is the reference for the next.

**Survive**  
No licence fee, pay-per-use cloud. Running cost fits in the service charge States already add to verification fees. No lock-in: public source code, standard Postgres, can move to State cloud.

**Get funded**  
SIH award and incubation first, then Startup India Seed Fund or MeitY Startup Hub grants to reach the pilot; State e-governance budgets pay for rollout as a work order.

## POTENTIAL CHALLENGES AND RISKS → STRATEGIES

|                          |                                                                           |
|--------------------------|---------------------------------------------------------------------------|
| No network at site       | Offline app; syncs, signs and notifies later                              |
| Fake / copied QR         | Signed QR + revocation list; serial shown on scan                         |
| Rules differ by State    | Fees and validity are data per State                                      |
| Data privacy (DPDP 2023) | Row-level security, minimal public fields                                 |
| AI mistakes              | Advisory only; officer decides; logged                                    |
| Officer adoption         | Field trial first; plain home screen per role; text size, contrast, Hindi |



**Complete live demo**  
Scan for the guided demo: register → apply → officer inspects (try offline) → signed certificate → scan the QR on another phone. Each step ticks itself.  
Evaluator password: TulaDemo@2026  
[tula-legal-metrology.vercel.app/demo](https://tula-legal-metrology.vercel.app/demo)



# IMPACT AND BENEFITS



## POTENTIAL IMPACT ON THE TARGET AUDIENCE

LMOs & GATCs

- Jobs on the phone, even offline
- Pass / fail calculated for them
- Certificate issued on site

Businesses

- Apply and pay without a visit
- SMS / email at every step
- Reminder before a stamp expires

Consumers

- Scan any certificate QR
- Fakes and edits are rejected
- Report a problem in two taps

Government

- Pending by stage, age, district
- Inspections due and overdue
- Full audit trail per instrument

## MEASURED ON THE LIVE SYSTEM (30 Sep 2026, scripted end-to-end run on the production database)

2 min 51 s

application filed → signed certificate (system time, 3 people, no paper)

6 s

certificate issued → signed by the server

6 / 6

forged or edited QR copies rejected (automated tests)

0

paper forms and office visits in the run

## PILOT TARGETS vs TODAY'S PAPER BASELINE (baseline measured from the pilot district's register)

≤ 7 days

application → certificate (median)

< 3 s

certificate check by QR scan

100%

due instruments reminded 30 days early

0

paper certificates in the pilot district

## BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

Social

Fair weight at shops, pumps and mandis. Buyers can check.

Economic

Fewer visits and faster renewals for traders and MSMEs.

Environmental

Paperless forms, certificates and records.

Governance

One auditable trail from application to enforcement.

## DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK



### References & Links

- LM Act 2009; LM (General) Rules 2011 · [consumeraffairs.gov.in](http://consumeraffairs.gov.in)
- LM (GATC) Rules 2013; State Enforcement Rules
- OIML R 76-1 (scales), R 117-1 (dispensers) · [oiml.org](http://oiml.org)
- IT Act 2000, DPDP Act 2023
- Supabase, Vercel and Gemini API docs

### COMPARED WITH TODAY

|                 | Paper          | Typical State portal | TULA               |
|-----------------|----------------|----------------------|--------------------|
| Certificate     | copyable paper | editable PDF         | signed QR          |
| Buyer can check | no             | rarely               | any phone, offline |
| Field record    | register       | typed later          | phone, offline     |
| Pass / fail     | judgement      | typed                | computed vs MPE    |
| Across States   | separate       | one per State        | one platform       |



**Scan: a real certificate**  
 DL/LM/2026/08913, issued and signed on the live system on 30 Sep 2026. Your phone checks the signature.  
 Code: [github.com/prerak2507/tula-legal-metrology](https://github.com/prerak2507/tula-legal-metrology)